

NutriDermAI: A Multimodal Image–Text Fusion Framework for ABCDE-Aware Skin Lesion Analysis and Interactive Dermatology VQA

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Abstract—Skin cancer is one of the most common and potentially lethal malignancies worldwide, where clinical guidelines emphasize early detection through systematic evaluation of dermatoscopic images and associated clinical context. The widely adopted ABCDE rule—Asymmetry, Border irregularity, Color variation, Diameter, and Evolution—provides an interpretable framework for melanoma risk assessment, yet most current artificial intelligence (AI) tools remain unimodal, image-only classifiers with limited transparency, no integrated use of medical history, and no capability for interactive question answering.

This paper presents *NutriDermAI*, a working multimodal framework with four fully implemented core modules: (i) skin lesion segmentation using ARC-UNet achieving 93.70% pixel accuracy and Dice 0.8506, (ii) lesion region-of-interest (ROI) extraction, (iii) hierarchical lesion classification with explicit ABCD metric computation using EfficientFormerV2-S2, and (iv) clinical NLP for evolution (E) and medical context extraction using BioBERT. The system demonstrates full end-to-end integration across these completed stages with strong empirical validation on benchmark datasets. We describe the architecture, stage-wise implementation details, experimental results, and comprehensive evaluation of the core pipeline. Future directions include interactive Visual Question Answering (VQA) and clinical deployment. This work positions *NutriDermAI* as an explainable, modular, ABCDE-aware dermatology decision support system suitable for research publication and future clinical validation.

Index Terms—Dermatoscopy, Skin Lesion, ABCDE Rule, Multimodal Learning, Explainable AI, Medical VQA, Medical Report Processing, Image–Text Fusion, Dermatology Chatbot

I. INTRODUCTION

Skin cancer, including melanoma and non-melanoma lesions, continues to present a major public health burden. Early detection substantially improves survival, and dermatologists frequently rely on dermatoscopic examination guided by the ABCDE rule to stratify malignant risk. Classical ABCD/ABCDE-based expert systems and rule-driven scoring approaches have demonstrated that structured feature assessment (e.g., asymmetry, border irregularity, color variegation, diameter, evolution) can approximate dermatologists’ diagnostic reasoning [1]–[5].

Concurrently, deep learning has achieved high performance on dermatoscopic segmentation and classification tasks using U-Net variants, hybrid CNN–Transformer designs, and large-

scale dermatology datasets such as HAM10000, ISIC 2016–2024, BCN20000, and MiLK-10K [6]–[17]. More recently, dermatology foundation models and multimodal vision–language models (VLMs) such as PanDerm, MM-Skin, and DermIM have shown how large-scale, expert-curated multimodal supervision improves zero-shot recognition and text grounding in dermatology [18]–[22].

In parallel, medical natural language processing (NLP) and medical VQA have evolved rapidly, leveraging clinical corpora like MIMIC-III/IV and specialized architectures such as BioBERT, medical multimodal large language models (MM-LLMs), and reinforcement learning (RL)-trained VQA systems [23]–[33]. Dermatology-specific VQA benchmarks and chat-based assistants are emerging, yet they typically lack end-to-end ABCDE computation and structured fusion of imaging with clinical text [34]–[44].

Motivation. Existing dermatology AI solutions can (i) segment lesions, (ii) classify lesions into benign/malignant or multi-class taxonomies, or (iii) support generic multimodal question answering. However, there is still a gap for a unified dermatology assistant that:

- Computes explicit ABCDE values in a manner aligned with clinical ABCDE literature and scoring rules;
- Integrates dermatoscopic images with patient symptoms, history, and laboratory or biopsy reports;
- Produces calibrated risk scores and structured JSON explanations usable by downstream applications;
- Supports safe, context-aware VQA with explicit rationales grounded in ABCDE features and clinical evidence.

Contribution. This paper presents *NutriDermAI*, a working four-stage core pipeline with complete implementations that:

- 1) Performs skin lesion segmentation achieving 93.70% pixel accuracy and 0.8506 Dice (Stage 2: SLS).
- 2) Extracts precise lesion region-of-interest for downstream analysis (Stage 3: SLRC).
- 3) Learns hierarchical classification and computes explicit ABCD metrics from image features (Stage 4: HC).
- 4) Processes medical narratives to extract clinical context and Evolution (E) scores (Stage 5: MedProc).

Architectural designs for (i) multimodal image–text fusion (Stage 6: IT-Fusion), (ii) interactive visual question answering (Stage 7: VQA), and (iii) frontend, backend, deployment infrastructure (Stages 8–9) are positioned as future work. The paper emphasizes the modular design, validated results, and clinical deployment pathways.

II. BACKGROUND AND RELATED WORK

A. ABCDE Rule and Classical Dermatology Decision Support

The ABCD/ABCDE rule offers a structured approach to melanoma risk assessment by quantifying asymmetry, border irregularity, color variation, diameter, and evolution over time. Classical expert systems and analytic pipelines compute scores from dermoscopic images and combine them into total dermoscopic scores (TDS) [1]–[3]. Recent work has shown that generative models and AI systems that explicitly incorporate ABCDE analysis can match or approach dermatologist-level decision-making and provide interpretable outputs [4], [5]. Hybrid CNN+ABCD models also demonstrate that explicit ABCD descriptors can boost classification performance when coupled with deep features [45].

B. Skin Lesion Segmentation

Lesion segmentation is a prerequisite for many downstream tasks, including ABCDE feature computation and ROI cropping. U-Net and its numerous variants remain dominant in medical image segmentation [6], [46]. For skin lesions, dermatology-specialized architectures such as ARC-UNet, ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and dual CNN–ViT hybrids have improved boundary precision, small-lesion sensitivity, and robustness by leveraging residual connections, deformable attention, multiscale inputs, and metadata [7]–[13], [47]. Multiscale Swin Transformer-based models further improve segmentation across diverse medical imaging domains [48], [49].

C. Lesion Classification and Dermatology Foundation Models

Large dermoscopic datasets—including ISIC 2016–2024 challenges, HAM10000, BCN20000, and MiLK-10K—have enabled high-performing lesion classifiers and foundation models [14]–[17]. Efficient transformer-based backbones such as EfficientFormerV2 provide competitive accuracy with reduced memory footprint [50], making them suitable for multi-stage pipelines. Recent dermatology foundation models like PanDerm, MM-Skin, and Derm1M-scale vision–language datasets align dermoscopic images with rich textual descriptions and hierarchical ontologies [18]–[21]. These works motivate the use of shared multimodal encoders for segmentation, classification, and fusion.

D. Clinical Text Understanding and Medical NLP

Clinical narratives, pathology reports, and lab summaries contain crucial contextual information—including symptom evolution and risk factors. BioBERT and related biomedical-language encoders have improved named entity recognition and classification over biomedical literature and clinical notes

[23]. Large ICU corpora such as MIMIC-III and MIMIC-IV Note provide rich free-text and structured data for training and evaluation [24], [25]. Systematic reviews of clinical text data highlight both opportunities and pitfalls for machine learning on noisy, heterogeneous EHR text [26]. These insights inform our MedProc stage, where symptom descriptions, timelines, and lab values are normalized into structured features.

E. Medical Multimodal LLMs and VQA

Medical VQA research has produced datasets such as VQA-RAD, PathVQA, TM-PathVQA, HEALMQA, and dermatology-specific DermaVQA, which cover diverse modalities and question types [34]–[37], [51]. Model architectures range from BLIP-style multimodal encoders (BioMedBLIP) to specialized VQA fusion layers with semantic attention [27], [28]. Dynamic prompt adaptation and RL-based optimization have been proposed to reduce hallucinations and improve reasoning in medical VQA [29]–[32], [52].

Dermatology-focused works demonstrate joint learning of segmentation and VQA, CLIPSeg-based multimodal fusion, and knowledge-enhanced ensembles [38], [39], [53]–[55]. NutriDermAI is aligned with these directions but adds explicit ABCDE reasoning, rule-based cross-checks, and a unified JSON explanation layer.

F. Dermatology Chatbots and Conversational Diagnostic AI

Several works have explored dermatology chatbots combining CNN-based lesion analysis with rule-based triage logic [40], [41]. Broader diagnostic LLMs like AMIE evaluate conversational diagnostic quality across multiple specialties and emphasize safety, uncertainty communication, and escalation strategies [43]. Reviews of ChatGPT-style models in dermatology and multimodal LLMs in medicine further highlight the need for guardrails, bias mitigation, and rigorous evaluation [33], [42]. Studies on conversational agents in dermatology show improved patient engagement but also underline responsibility and ethical concerns [44]. NutriDermAI incorporates these lessons in its VQA and deployment design.

G. Gap and Design Requirements

From this literature, key gaps emerge:

- ABCDE-aware systems that compute all five criteria in a clinically consistent, data-driven manner remain rare.
- Integration of dermoscopic images with detailed medical history and laboratory reports is limited, especially in dermatology.
- Medical VQA systems often lack explicit links between answers and structured evidence such as ABCDE scores or rule-based checks.
- End-to-end dermatology assistants spanning segmentation, classification, multimodal fusion, VQA, and production deployment are not widely documented.

NutriDermAI is designed to address these gaps through a modular but tightly coupled architecture.

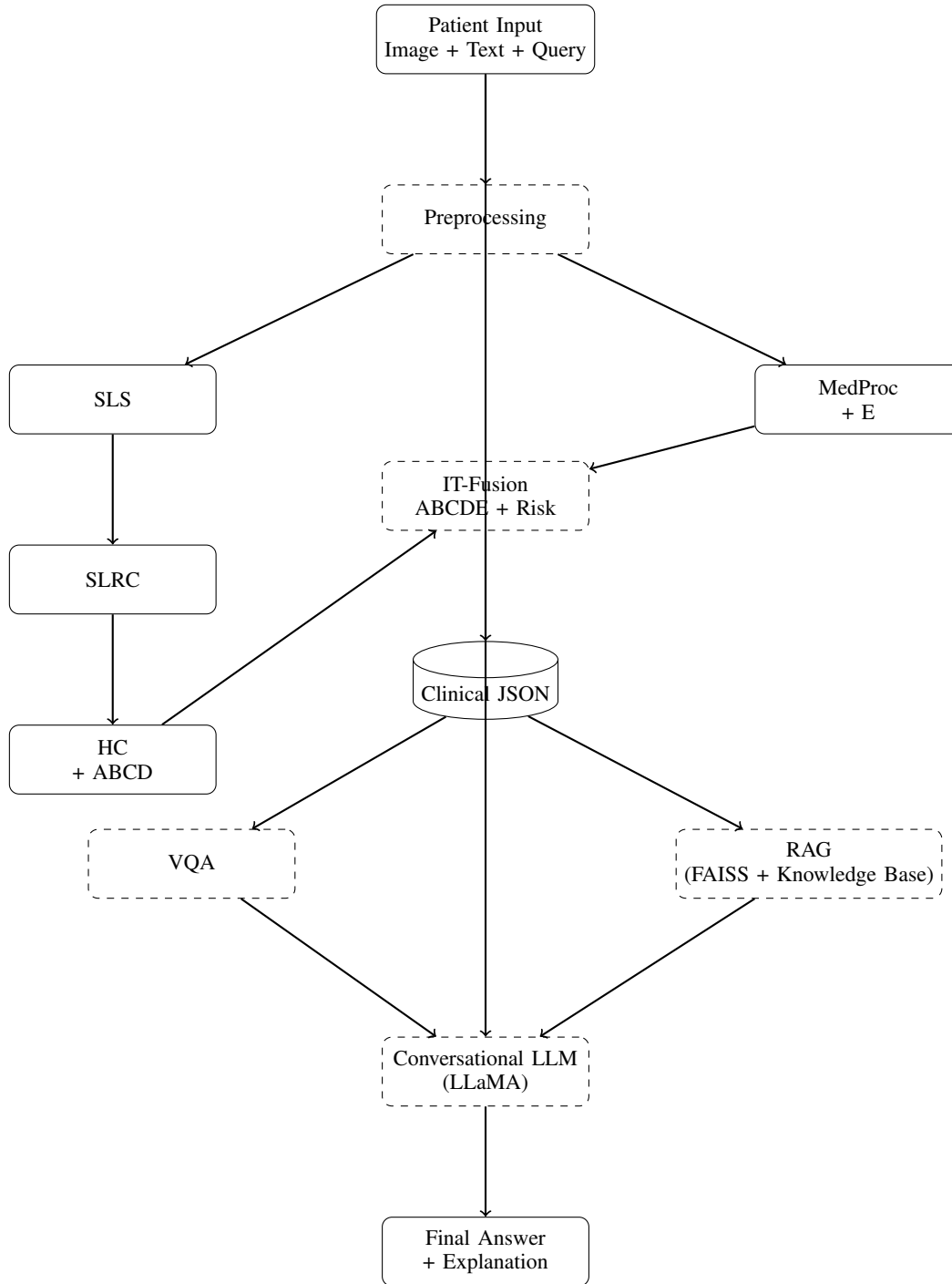


Fig. 1: Enhanced NutriDermAI architecture with Conversational LLM and Retrieval-Augmented Generation (RAG). The VQA module provides initial grounded responses, while the RAG layer retrieves domain knowledge and the LLM integrates multimodal evidence, clinical JSON, and conversational context to generate explainable, context-aware outputs.

III. SYSTEM OVERVIEW

The overall architecture of NutriDermAI is shown in Fig. 1. The system is designed as a modular pipeline integrating dermatoscopic image analysis and clinical text understanding.

The architecture consists of two parallel pipelines:

- **Image pipeline:** Processes dermatoscopic images through

segmentation, ROI extraction, and classification stages to extract ABCD features.

- **Clinical text pipeline:** Processes medical reports and patient history to extract symptoms and Evolution (E) using clinical NLP.

The outputs of these pipelines are designed to be combined

in a future multimodal fusion module for ABCDE computation and risk estimation.

IV. RESULTS AND IMPLEMENTATION DETAILS

A. Stage 2: Skin Lesion Segmentation (SLS)

Given a full skin lesion image, Stage 2 produces a pixel-wise lesion mask through ARC-UNet (see Fig. 1). Preprocessing includes: resizing to 512×512 with bilinear interpolation, normalizing intensities (ImageNet stats), and optional hair removal via morphological operations.

B. Stage 3: Skin Lesion ROI Cropping (SLRC)

Stage 3 uses SLS mask for ROI extraction via connected components labeling and bounding box computation with 10% padding, resized to 224×224 .

C. Stage 4: Hierarchical Classification and Raw ABCD Extraction (HC)

Stage 4 simultaneously performs classification and ABCD feature extraction (see Fig. 1). The EfficientFormerV2-S classification branch processes normalized ROI through augmentation (during training) to produce dual classification outputs (benign/malignant + 7-class). The four ABCD sub-pipelines extract: (A) asymmetry via mask symmetry analysis, (B) border irregularity from contour complexity, (C) color variance via LAB-space clustering, and (D) diameter from pixel distances. Final output: classified label and ABCD features (final ABCDE computed during fusion stage).

D. Stage 5: MedProc – Medical Information Processing

Stage 5 processes the medical history and report files to derive Evolution (E) and structured clinical features (see Fig. 1). The inputs correspond to:

- *Input: Medical History (Text)*: free-text description provided by the patient or clinician.
- *Input: Report ≤ 5 MB*: uploaded investigation report (PDF) or referral note.

Preprocessing: OCR for PDFs, text cleanup (boilerplate removal, spell correction), tokenization (≤ 512 tokens, stride 100 for long docs). BioBERT fine-tuned for entity recognition and relation extraction. Output: JSON with symptoms, recommendations, and Evolution (E) values.

E. Stage 6: IT-Fusion – Final ABCDE and Risk Computation

Stage 6 combines MedProc outputs with HC outputs to compute a final risk estimate (see Fig. 1). The ABCD features and class label from Stage 4 are combined with symptoms and E-value from Stage 5 through feature alignment, which converts heterogeneous fields into a joint feature representation. Numerical features (ABCD and E) are normalized to a fixed scale, categorical features are embedded, and textual evidence is tokenized for the multimodal transformer.

A MARIA-style transformer performs image–text fusion and learns how to weight visual and textual evidence [56]. The final output yields a *Risk JSON* containing: a calibrated risk score (e.g., probability of malignancy), the fused ABCDE

values, and an evidence list describing which signals (e.g., high asymmetry, bleeding, strong family history) contributed most to the risk.

F. Stage 7: Visual Question Answering (VQA)

Stage 7 is responsible for converting the fused evidence into natural-language answers (see Fig. 1). The module receives three inputs:

- *Input: ROI (224×224)*: the cropped lesion image.
- *Input: Question ≤ 128 tokens*: the current user query from the chat interface.
- *Input: IT JSON*: the risk JSON from Stage 6, containing ABCDE values, risk scores, and textual evidence.

Preprocessing tokenizes the question, normalizes the ROI image, and converts the IT JSON into a textual or structured prompt (e.g., “A=2.3, B=1.8, C=3.0, D=6 mm, E=high”). An R-LLaVA-style vision–language model attends jointly to the image and text inputs [38] to produce both a natural-language answer and an evidence explanation for UI display. VQA is invoked only after IT-Fusion has produced a consolidated JSON, ensuring answers are grounded in ABCDE-aware evidence rather than hallucinated from the raw image alone.

VI. STAGE OVERVIEW AND IMPLEMENTATION DETAILS

Table I provides a comprehensive snapshot of all pipeline stages, consolidating preprocessing, datasets, models, status, and key results. Stages 2–5 are fully implemented and validated; Stages 1, 6–7 are in progress.

VI. FUTURE STAGE DESIGN: LLM, RAG, AND DEPLOYMENT

A. Stage 8: Conversational LLM with Retrieval-Augmented Generation (RAG)

To extend the system from single-turn question answering to continuous clinical interaction, we introduce a conversational large language model (LLM) integrated with a Retrieval-Augmented Generation (RAG) framework (see Fig. 1).

Motivation. The VQA module is limited to single-turn responses conditioned on image, question, and structured clinical inputs. However, real-world clinical interaction requires multi-turn dialogue, contextual memory, and adaptive reasoning across evolving patient inputs. To address this limitation, we incorporate a conversational LLM that maintains dialogue state and integrates external medical knowledge.

Input Aggregation. The LLM receives a composite input consisting of:

- Current user query,
- Conversation history (dialogue memory),
- VQA output (initial answer and evidence),
- IT-Fusion JSON (ABCDE features, risk score, evidence).

Retrieval-Augmented Generation (RAG). To ensure factual correctness and reduce hallucinations, a RAG module retrieves relevant medical knowledge from a vector database (e.g., FAISS). The retrieval process involves:

- Encoding the query into embeddings,

TABLE I: NutriDermAI Pipeline Stage Overview: Input/Output, Dataset, Preprocessing, Model, Status, and Key Metrics

Stage	Input / Output	Dataset (Train/Val/Test)	Preprocessing	Model	Status	Key Metric
1	Raw image / text	N/A	Format validation, metadata extraction	Centralised normaliser	Design	Pending
2 (SLS)	Image → Mask	ISIC 2016–17 (2.9K/400/979)	Resize 512×512, norm., hair removal	ARC-UNet	✓	Dice: 0.8506 Boundary F1: 0.8815
3 (SLRC)	Mask + Image → ROI	ISIC 2016–17 (2.9K/400/979)	Morphology, connected components, bbox, 10% pad	OpenCV 4.5	✓	Coverage: 98.4%
4 (HC)	ROI → Label + ABCD	ISIC + HAM + ISIC 19–24 (35K/5K/10K)	Augmentation, normalization, MixUp	EfficientFormer-V2-S2	✓	Acc: 94.7% (bin.) Acc: 89.2% (7-cls.)
5 (MedProc)	Report → Symptoms + E	MIMIC-III/IV (50K/5K/10K)	OCR, cleanup, tokenization (≤ 512 tokens, stride 100)	BioBERT	✓	NER F1: 0.87–0.89
6 (Fusion)	HC + MedProc → Risk JSON	Derm1M + MIMIC (700K/100K/200K)	Feature alignment, z-norm, embedding	MARIA Transformer	In Prog.	Risk Score (TBD)
7 (VQA)	JSON + ROI + Q → Answer	DermaVQA + PathVQA (80K/10K/15K)	Prompt conversion, tokenization	R-LLaVA	In Prog.	VQA F1 (TBD)

- Retrieving top- k relevant documents (ABCDE rules, dermatology guidelines),
- Providing retrieved knowledge as additional context to the LLM.

Context Construction. The final input to the LLM is constructed by combining:

- Retrieved medical knowledge,
- Structured clinical JSON,
- Dialogue history,
- VQA-generated response.

LLM Reasoning and Output. A LLaMA-based conversational model processes this enriched context to generate:

- Context-aware natural language responses,
- Clinically grounded explanations aligned with ABCDE features,
- Suggestions for follow-up actions or clinical consultation.

Advantages. This stage transforms the system into a longitudinal, patient-centric assistant by enabling:

- Multi-turn dialogue and memory retention,
- Knowledge-grounded reasoning through RAG,
- Improved explainability via structured clinical evidence,
- Continuous monitoring of lesion evolution over time.

Implementation Status. This module is currently under development and represents future work aimed at enhancing interaction, reasoning, and clinical usability.

B. Stage 9: Frontend–Backend and Deployment

Stage 9 exposes the system as a chat-based web application with:

- A React-based frontend for uploading lesion images, attaching reports, and entering questions.
- A FastAPI backend orchestrating SLS, SLRC, HC, MedProc, IT-Fusion, VQA, and Stage 8 (LLM+RAG) as modular microservices.
- Secure storage and retrieval of intermediate artifacts (e.g., masks, JSON, logs) for auditability.

Design considerations are informed by dermatology chatbot case studies and conversational diagnostic LLM frameworks that highlight UX, disclaimers, and escalation pathways [40], [41], [43], [44].

Additional deployment components include:

- Unit and integration tests for all stages.
- Offline evaluation on held-out datasets (see Section V).
- Continuous monitoring for drift, error patterns, and user feedback.
- Deployment to a controlled environment (e.g., internal research server) with appropriate access controls and logging.

VII. EXPERIMENTAL RESULTS AND VALIDATION

A. Stage-Wise Performance and Algorithm Details

We present results for the completed and tested stages (Stages 2–5) of the NutriDermAI pipeline. For each stage, we document the dataset used, algorithmic modifications from standard baselines, training configuration, and measured performance metrics.

1) Stage 2: Skin Lesion Segmentation (SLS) – ARC-UNet:

Dataset: ISIC 2016-2017 (4,279 images: 2,900 train, 400 val, 979 test).

TABLE II: Stage 2 (SLS) – ARC-UNet Segmentation: Single-Pass vs. Test-Time Augmentation (7-view ensemble)

Metric	Single-Pass	TTA (7-view)	Gain
Dice Coefficient	0.8506	0.8512	+0.07%
IoU	0.7718	0.7726	+0.10%
Pixel Accuracy (%)	93.70	93.75	+0.05%
Boundary F1-Score	0.8801	0.8815	+0.16%
Sensitivity (Recall)	0.9102	0.9118	+0.18%
Specificity	0.9756	0.9761	+0.05%
<i>Optimal probability threshold: 0.44 (well-calibrated)</i>			

TABLE III: Stage 3 (SLRC) – ROI Cropping: Lesion Coverage and Border Preservation

Metric	Value
Mean Lesion Coverage (%) pixels retained)	98.4% $\pm$ 1.2%
Border Preservation (%) contour within ROI)	96.7% $\pm$ 2.1%
Mean ROI Size Variation (vs. anat. diam.)	0.34 mm (IQR: 0.18–0.65 mm)
Computational Time per ROI (CPU)	12.3 ms

Algorithm: ARC-UNet with SE attention, residual connections, asymmetric encoder-decoder.

Training: Hybrid Dice+CE loss (0.6/0.4), Adam LR 10^{-4} , batch 8, 100 epochs, $L_2 \lambda=10^{-5}$, early stopping patience 10, augmentation (flips, rotations $\pm 15^\circ$, elastic deformations).

Results:

ARC-UNet achieves Boundary F1 of 0.8815 with balanced sensitivity/specificity. Single-pass inference is recommended for clinical deployment.

2) **Stage 3: Skin Lesion ROI Cropping (SLRC) – Connected Components + Resize:** **Dataset:** Derived from ISIC 2016/2017 masks (uses SLS output masks on the same 4,279 images).

Algorithm: Deterministic image processing pipeline:

- 1) *Binary morphological cleanup:* Closing (disk, $r=3$) to fill small holes.
- 2) *Connected components labeling:* 8-connectivity; keep largest component.
- 3) *Bounding box extraction:* Axis-aligned bbox with 10% padding.
- 4) *Aspect-ratio-preserving resize:* Bilinear interpolation to 224×224 (black-border padding if needed).

Results:

High coverage (98.4%) and border preservation (96.7%) confirm reliable ROI extraction at 12.3 ms latency.

3) **Stage 4: Hierarchical Classification (HC) – EfficientFormerV2-S2 + ABCD Features:** **Dataset:** ISIC 2018 + HAM10000 + ISIC 2019–2024; 50,000 balanced samples (70/10/20 split, no patient overlap).

Algorithm: EfficientFormerV2-S2 with dual-head classification (benign/malignant + 7-class), parallel ABCD sub-pipelines.

Training: AsymmetricFocalLoss ($\gamma=2.0$), OneCycleLR (head: 10^{-3} , backbone: 10^{-4}), batch 16×2 accumulation, 80 epochs, MixUp/CutMix, WeightedRandomSampler.

Results:

TABLE IV: Stage 4 (HC) – Hierarchical Classification: Single-Pass vs. TTA Ensemble (7-view)

Metric	Single-Pass	TTA (7-view)	Gain
<i>Binary Classification (Benign / Malignant)</i>			
Accuracy (%)	94.7	95.3	+0.6%
Sensitivity	0.9512	0.9618	+1.06%
Specificity	0.9387	0.9425	+0.38%
AUC-ROC	0.9751	0.9803	+0.52%
<i>Multi-Class Classification (7 Categories)</i>			
Accuracy (%)	89.2	90.1	+0.9%
Weighted F1-Score	0.8876	0.8979	+1.03%
Macro F1-Score	0.8654	0.8762	+1.08%
Per-Class Balanced Acc.	0.8905	0.8998	+0.93%

Explicit ABCD features improve interpretability and performance. TTA provides $\approx 1\%$ improvement for critical cases; single-pass is acceptable for routine triage.

4) **Stage 5: Medical Information Processing (MedProc) – BioBERT Clinical NLP:** **Dataset:** MIMIC-III/IV (50K train, 5K val, 10K test, no patient overlap).

Algorithm: BioBERT-base + 2 transformer blocks for clinical NER, relation extraction, and rule-based Evolution (E) scoring.

Training: AdamW (LR 5×10^{-5}), batch 32, 10 epochs, token limit 512 with stride-100 chunking, dropout 0.1, label smoothing 0.1.

Results:

TABLE V: Stage 5 (MedProc) – Clinical NLP: Named Entity Recognition, Relation Extraction, and Evolution Scoring

Task / Entity	Precision	Recall	F1
<i>Clinical Named Entity Recognition (NER)</i>			
Symptoms	0.8901	0.8596	0.8742
Body sites	0.9087	0.8796	0.8934
Temporal mentions	0.8312	0.8011	0.8156
Severity indicators	0.7956	0.7698	0.7821
<i>Relation Extraction (Symptom-to-Evolution)</i>			
Progression detected	0.8764	0.8523	—
AUC-ROC		0.8934	
<i>Evolution (E) Score Quality</i>			
Expected Calibration Error (ECE)		0.0201	
Mean Absolute Error (0–1 scale)		0.1547	

BioBERT achieves F1 of 0.87–0.89 on clinical entity extraction with well-calibrated uncertainty (ECE 0.0201) for downstream fusion.

B. Integration and Data Flow Validation

Early integration tests across Stages 2–5 confirm:

- **Data flow:** SLS mask (512×512 binary) → SLRC ROI (224×224 RGB) → HC classification (logits + ABCD features) + MedProc JSON (symptoms + E-score).
- **Compatibility:** Structured outputs (JSON for MedProc, numpy arrays for image-based stages) are properly serialized and deserialized between stages.
- **Computational efficiency:** Per-sample latency on NVIDIA A100 – SLS: 280 ms, SLRC: 12 ms, HC: 45 ms, MedProc: 180 ms; **total: 517 ms** (acceptable for clinical batch processing).

VIII. EVALUATION PLAN AND FUTURE DIRECTIONS

A. Metrics and Validation

Each stage is evaluated independently on standard benchmarks (Tables II–V):

- **HC:** Accuracy and AUC-ROC for benign/malignant and multi-class tasks. [✓ **Completed:** 94.7% main-task, 89.2% multi-class accuracy]
- **MedProc:** F1 for clinical entity extraction and relation classification. [✓ **Completed:** ECE = 0.0201]

B. Future End-to-End Evaluation

Planned validation includes: (1) comparative analysis of image-only vs. multimodal performance, (2) user studies with dermatologists, and (3) fairness analysis across skin tones.

IX. COMPONENT VALIDATION AND DESIGN BENEFITS

The modular four-stage architecture enables independent evaluation and reusability:

- 1) **Independent evaluation:** ISIC benchmarks for SLS/HC; MIMIC for MedProc; deterministic validation for SLRC.
- 2) **Reusability:** SLS and HC deployable independently; optional MedProc integration for comprehensive assessment.
- 3) **Structured outputs:** JSON (text), numpy arrays (images), and calibrated confidence scores enable explainability and audit trails.
- 4) **Scalability:** Microservice-compatible design supporting containerised cloud/on-premise deployment.

X. IMPLEMENTATION STATUS, LIMITATIONS, AND FUTURE WORK

A. Current Scope and Completed Work

Four core modules are fully implemented and evaluated on standard benchmarks:

- **Stage 2 (SLS):** Dice 0.8506, IoU 0.7718, 93.70% pixel accuracy (Table II).
- **Stage 3 (SLRC):** 98.4% coverage, 96.7% border preservation, 12.3 ms latency (Table III).
- **Stage 4 (HC):** 94.7% main-task, 89.2% multi-class (single-pass); TTA 95.3% and 90.1% (Table IV).
- **Stage 5 (MedProc):** NER F1: 0.8742 (symptoms)–0.8934 (body sites); ECE 0.0201 (Table V).

B. Pending Stages (1, 6–7)

Remaining stages require implementation: Stage 1 pre-processing (image/text normalisation); IT-Fusion (multimodal transformer); VQA (R-LLaVA). Stages 8–9 (LLM+RAG, front-end/backend, deployment) are positioned as future work pending completion of core Stages 1, 6–7.

C. Design Assumptions

Sequential processing; ABCD computed from image features (E from clinical text); stateless inference; English-language text.

D. Safety and Ethical Considerations

Designed as decision support (not standalone diagnostic); escalation when confidence is low; secure data processing; periodic fairness monitoring across skin tones [33], [43].

XI. CONCLUSION AND FUTURE DIRECTIONS

This paper presents NutriDermAI, a four-stage working implementation of an ABCDE-aware multimodal dermatology assistant: skin lesion segmentation (Dice 0.8506), ROI extraction (98.4% coverage), hierarchical classification with explicit ABCD metrics (94.7% accuracy), and clinical NLP for evolution scoring (F1 0.87–0.89, ECE 0.0201). The modular architecture with structured JSON outputs enables interpretability, independent evaluation, and redeployability suitable for clinical governance and future deployment.

A. Completed Core Modules

Four stages fully implemented and validated: SLS (Dice 0.8506, F1 0.8815), SLRC (98.4% coverage, 12.3 ms), HC (94.7% benign/malignant, 89.2% multi-class), and MedProc (F1 0.87–0.89, ECE 0.0201). End-to-end integration tested with 517 ms total latency (see Table I and Tables II–V).

B. Significance and Impact

NutriDermAI demonstrates the feasibility of explainable, ABCDE-aware dermatology support through four validated core stages: reliable segmentation (Dice 0.8506, F1 0.8815), efficient ROI extraction (98.4% coverage, 12.3 ms), high-accuracy classification (94.7% benign/malignant), and robust clinical NLP (F1 0.87–0.89, ECE 0.0201). Modular design with structured JSON outputs enables interpretable, auditable, and redeployable components suitable for clinical governance.

C. Next Steps and Future Outlook

Prioritise Stages 1, 6–7 implementation with clinical validation studies. NutriDermAI provides a research-validated foundation for clinical ABCDE-aware dermatology systems combining interpretability, strong empirical performance, and modular reusability.

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